EHR + DIABETES DIGITAL TWIN

A conversation becomes evidence. Evidence powers a twin.

One continuous patient journey - from clinical capture to explainable what-if exploration.

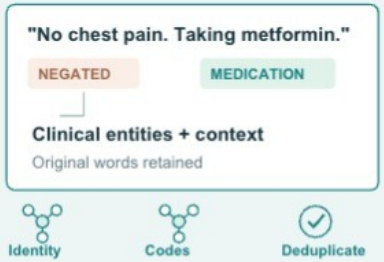
EHR / CAPTURE → UNDERSTAND → VERIFY → PRESERVE

1 Capture the encounter



Speech · reports · lab APIs
STT / OCR supplied upstream

2 Give the words meaning



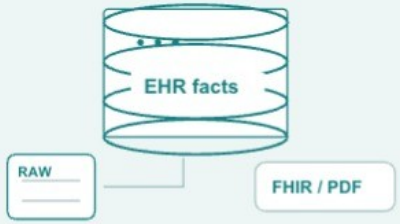
Match patient → fuse → extract
Terminology + source provenance

3 Doctor verifies the facts



Review context, uncertainty and conflicts
Corrections + approval leave an audit trail
Similar-case suggestions remain optional

4 Preserve a trusted record

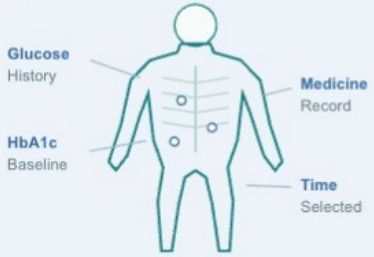


PostgreSQL facts · MinIO raw sources
FHIR resources + clinical documents

Encounter-to-cohort identity link is pending in this prototype

DIGITAL TWIN / CONTEXT → PREDICT → EXPERIMENT → EXPLAIN

5 Assemble the patient twin



Patient + selected time → past-only features
Demo: cohort CSVs + historical CGM
New reviewed facts do not retrain the model

6 Predict & explain the drivers



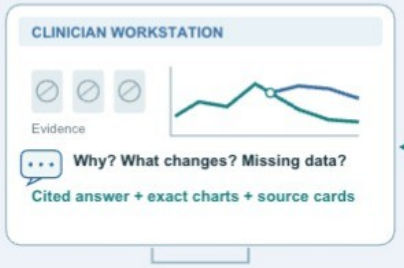
Spike risk + 15–120 minute trajectory
TreeSHAP shows mathematical drivers

7 Explore a different scenario



Compare curves + estimated differences
Sensitivity rules; hypothetical effects

8 Turn evidence into insight



Patient-scoped retrieval + backend tools
Groq / OpenRouter explain; clinician decides

Assistant flow Retrieve patient evidence → run supported tools → check values & citations → show the explanation

Patient-scoped access | **Original sources + review audit** | **Server-only provider keys** | **No autonomous prescribing**

Implemented twin path
EHR reference / integration pending
Historical data · not live monitoring · research prototype